

MSc Research Project

**Predicting and controlling the growth shape of crystals  
– A solvent selection approach**

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# Predicting and controlling the growth shape of crystals – A solvent selection approach

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**Abstract:** Predicting and controlling the crystal shape is critical for optimising industrial crystallisation processes, as the crystal shape directly influences downstream processing, product quality, and operational efficiency. This exploratory study introduces an integrated modelling framework that combines a mechanistic crystal growth model with a process-level optimisation model. The modelling framework allows prediction of crystal shapes and supports selection of solvents and process conditions to maximise yield, minimise solvent use, and avoid undesired crystal shapes. Case studies on the crystallisation of biphenyl and naphthalene were considered to demonstrate the flexibility and broad applicability of the framework. The framework predicts the shape of biphenyl and naphthalene crystals over a range of different solvents and temperatures. The results of shape prediction exhibit a margin of error due to the approximations and simplifications adopted to make the mechanistic model computationally simple to demonstrate feasibility of the approach. Optimisation of Biphenyl and Naphthalene crystallisation processes allow to identify optimal solvents and process conditions. Toluene was identified as an optimal solvent for Biphenyl crystallisation at  $T = 308$  K while for Naphthalene crystallisation, Acetone was suggested as the optimal solvent at  $T = 306$  K by the framework.

**Keywords:** *Crystallisation, crystal shape, solvent selection, optimisation*

## □ Introduction

Solvent-based crystallisation is one of the most widely used industrial processes for the separation and purification of materials. It is used extensively during the production of pharmaceuticals, speciality chemicals, food ingredients, and advanced materials [1]. The end-use application of these crystalline products is highly dependent on their properties, such as crystal structure, crystal shape, and crystal size distribution. Crystal shape is of particular importance because it greatly influences the downstream process operations, such as filtration, drying, milling, storage and transport [2].

In the process industry, during crystallisation, a desired crystal structure (polymorph) is selected, and its seed is added to a crystalliser. The crystal grows and retains the crystal structure of the polymorph seed [3]. However, the shape of the resulting crystal can vary significantly depending on the growth conditions. Some of the crystal shapes may be undesirable for downstream processing. For example, needle-shaped crystals and plate-shaped crystals are fragile and tend to form compacted filter cakes that restrict flow. They also exhibit poor flow properties in hoppers and feeders and have low bulk densities that make packaging and transport inefficient [4], [5]. These problems translate into longer filtration times, higher energy costs, and potentially lower yields. On the other hand, controlling the growth conditions can produce block-shaped (equant) crystals [6]. Block-shaped crystals resist breakage, flow easily, pack efficiently, and filter without major blockages, making them highly desirable for downstream processing. However, for many compounds, needle shape or plate shape may correspond to the thermodynamically most favoured shape [7]. The functionality of the needle or plate shaped crystals can make them desirable in certain applications. In pharmaceuticals, needle shapes of

certain compounds can be beneficial because their large surface area accelerates dissolution, supporting faster bioavailability [8]. In advanced functional materials, specific aspect ratios are sometimes crucial for achieving optimal optical, mechanical, or electronic performance. The challenge, therefore, is not only to avoid certain problematic crystal shapes but also to selectively tailor shapes that meet different performance criteria in specific applications.

The solvent used for crystal growth and the amount (in moles) of solvent added to the crystalliser are critical variables responsible for altering the growth shape of crystals. The temperature of the crystalliser is also a factor which affects the kinetics of crystal growth, and which can be considered as an important variable [7]. Various mechanistic, non-mechanistic models account for the changes in growth conditions to predict the growth shape of crystals. However, they rarely provide information regarding process-level parameters, such as yield or solvent consumption. For the industry, shape prediction by itself is insufficient because process economics and sustainability are inseparable from crystal shape issues [1]. This has left many industrial practices dependent on empirical, trial-and-error solvent screening instead of predictive design.

To overcome this gap, Lovette *et al.* [6] have provided a causality-based guide for solvent selection to predict the formation of needle-shaped crystals. Whilst this proves to be a reliable guide to solvent selection, the process is not modelled directly in the approach, limiting its application.

In this study, a modelling framework is proposed to integrate mechanistic crystal growth model with broader process considerations. The novelty of this framework lies in its ability to

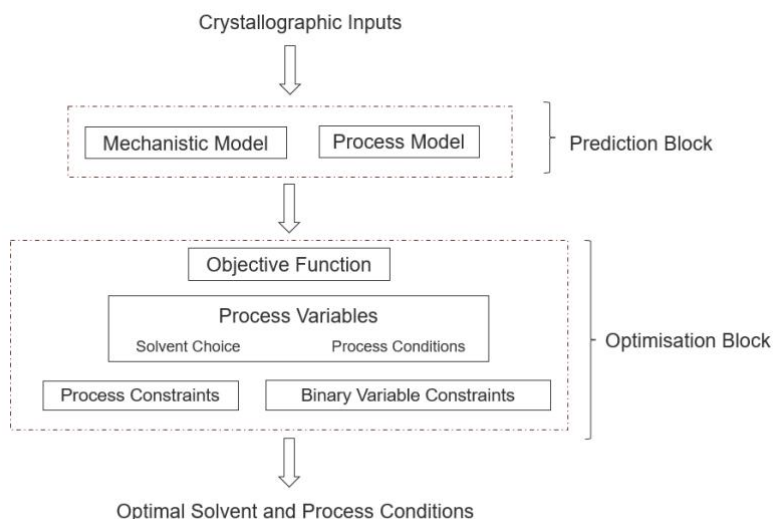


Figure 1 - Structure of the comprehensive modelling framework for crystal shape prediction and control

comprehensively evaluate both solvent consumption and process yield while predicting crystal shape at any given temperature in a specific solvent. Additionally, it offers optimisation capabilities for achieving desirable crystal shapes while ensuring high yields and low solvent consumption. In cases where the formation of needle-shaped crystals is unavoidable, the framework can suggest the addition of a milling step to the process design as a control measure. The framework distinguishes itself by making solvent selection a key decision variable in the optimisation process, recognising its significant impact on solubility, supersaturation, and interfacial interactions. This approach makes solvent choice a critical lever for shape prediction and control. Moreover, with global regulations urging a reduction in hazardous solvent use, selecting greener solvent alternatives for processes is not just environmental responsibility but also an industrial necessity [8].

## □ Methodology

Crystal growth in solution is a complex process in which the final shape of the crystals depends on multiple interconnected factors. The chemical potential difference ( $\Delta\mu$ ) between the growth environment (solution) and the crystal lattice acts as the driving force for the crystallisation process. It can be defined as a function of supersaturation ( $\sigma$ ) as [9]:

$$\Delta\mu = kT \ln(1 + \sigma) \quad (1)$$

The dimensionless term supersaturation ( $\sigma$ ) can be defined as the ratio of the solute concentration ( $x$ ) above the solubility limit to the solubility limit ( $x^{sat}$ ) [10].

$$\sigma = \left( \frac{x - x^{sat}}{x^{sat}} \right) \quad (2)$$

The solubility of any molecule in a selected solvent varies significantly with temperature - generally, higher temperatures allow more material to dissolve, while lower temperatures reduce solubility. Thus, the supersaturation is largely affected by solvent choice, the amount of solvent added and the temperature at which the crystallisation process is carried out. Therefore, accurate prediction and control of crystal shape requires careful consideration of three key factors: the solvent used, the amount of solvent added, and the system's temperature. The structure of the comprehensive modelling framework which accounts for all these factors is presented in Figure 1.

## Prediction Block

### *Mechanistic Model*

Various non-mechanistic models, such as the Bravais, Friedel, Donnay, and Harker (BFDH) model and the Attachment Energy (AE) model, are commonly used to predict crystal shape. However, these models are helpful in predicting a single crystal habit while ignoring the effect of growth conditions, like supersaturation [11], [12]. Monte Carlo simulations have demonstrated high accuracy in predicting crystal morphology; however, their extensive computational requirements make them largely impractical for routine industrial applications. Therefore, there is a need for a model that can capture the effects of growth conditions on crystal shape and provide rapid shape predictions. One such mechanistic modelling framework, which was developed by Doherty and co-workers [10] has proven to be successful in accurately predicting the shape of crystalline compounds. An adaptation of this mechanistic model is presented below.

At the base of mechanistic modelling, there is a need to determine the normal growth rate of a face  $i$ . The normal growth rate ( $G_i$ ) can be expressed as a function of step height  $h_i$  (an elementary step equal to interplanar spacing) and the time taken to advance the face by one layer ( $\tau_i$ ) [10].

$$G_i = \left( \frac{h_i}{\tau_i} \right) \quad (3)$$

The Frank-Chernov conditions (Equation 4) provide a relationship between the Growth rate ( $G_i$ ) and the distance of the face from the centre ( $H_i$ ). It states that for a given growth rate, there exists a steady state morphology that a crystal will evolve into, irrespective of its initial seed shape [13], [14], [15].

$$\left( \frac{G_1}{H_1} \right) = \left( \frac{G_2}{H_2} \right) \dots = \left( \frac{G_i}{H_i} \right) = \text{constant} \quad (4)$$

The relative growth rate ( $R_i$ ) is given as the ratio of the growth rate of a face with respect to a reference face [10].

$$R_i = \left( \frac{G_i}{G_{hkl}} \right) \quad (5)$$

The aspect ratio for a crystal can be defined using this relative growth rate as the ratio of the maximum ( $R_i$ ) to the minimum ( $R_i$ ). On calculating the growth rate for different crystal faces, one can obtain the value of relative growth rate. Calculating  $R_i$  can allow us to predict the aspect ratio of the crystal which corresponds to its shape. Further, in this study, the term “shape of crystal” refers to the aspect ratio or relative growth rate of crystal faces. The crystallography is the primary input for the model, and thus, the predictions are made for the desired polymorph seed. Therefore, the growth rate for a crystal face and thus the aspect ratio depends solely on  $\tau_i$  and not the step height  $h_i$  ( $h_i$  will remain constant even though the growth conditions change) [10].  $\tau_i$  depends on the mechanism and corresponds to the spiral rotation time for spiral growth and the face coverage time for 2D nucleation growth [16], [17].

In the presence of a screw dislocation on the surface of a crystal face, there is a continuous source of steps for the spiral edge as the initial step at the dislocation advances. This process continues until the initial edge is one layer higher than the starting point [16], [18]. The time taken to complete this full rotation is known as spiral rotation time. The growth rate for a face growing through a spiral mechanism can be written as:

$$G_s(i) = \left( \frac{h_i}{\tau_s} \right) \quad (6)$$

Spiral rotation time ( $\tau_s$ ) can be given as a function of  $\tau_s^*$  which is independent of  $\sigma$  (supersaturation) and  $k_0$  (Overall rate constant).

$$\tau_{s,i} = \frac{h_i \sigma \ln(1 + \sigma) k_0}{\tau_{s,i}^*} \quad (7)$$

$\tau_s^*$  is applicable to any supersaturation however its value depends on the solvent and temperature [10].

$$\tau_s^* = \sum_{i=1}^n \frac{l_{c,i} \ln(1 + \sigma) \sin(\alpha_i)}{a_{p,i} \rho_{k,i}} \quad (8)$$

$l_{c,i}$  is the critical length and  $\alpha_i$  is the angle between the edges of the crystal.  $a_{p,i}$  is defined as the step propagation length normal to the edge in the direction of a new row.  $\rho_{k,i}$  gives the distribution of kink sites on the crystal surface (also known as kink density). This distribution is given at thermodynamic equilibrium by minimisation of Helmholtz Free Energy over the edge [19].

$$\rho_{k,i} = \frac{2 \exp \left( \frac{-\phi_{k,i}}{k_B T} \right)}{1 + 2 \exp \left( \frac{-\phi_{k,i}}{k_B T} \right)} \quad (9)$$

The overall rate constant  $k_0$  can be written as:

$$k_0 = k_T^+ \exp \left( - \frac{\Delta W_{K,i} + \Delta W_{E,i} + \Delta W_T}{k_B T} \right) \quad (10)$$

$k_T^+$  is the rate constant for terrace adsorption and it is determined by the transition state barrier for desolvation in the terrace direction ( $\Delta U_T$ ), frequency pre-factor ( $\nu_0$ ), and temperature dependence. The desolvation energy barrier refers to the energy barrier that growth units must overcome when transitioning from solution phase to adsorbed positions on crystal terraces [9], [20].

$$k_T^+ = \nu_0 \exp \left( - \frac{\Delta U_T}{k_B T} \right) \quad (11)$$

For centrosymmetric crystals, this rate constant can be approximated as uniform across all the faces, simplifying kinetic analysis.

In Equation 10,  $\Delta W_{K,i}$  is the work done for detachment in the kink direction and  $\Delta W_{E,i}$ ,  $\Delta W_T$  are the detachment barriers from edge adsorbed to terrace-adsorbed and terrace adsorbed to solution respectively. For layered crystal growth in reasonable temperature and supersaturation ranges, the kink incorporation is the rate determining step with highest energy barrier. This makes the kink sites the most stable growth positions [21]. Thus, work done for detachment in the kink direction is significantly higher as compared work done for detachment in the edge and terrace direction. Therefore, for calculation of  $k_0$ , the effect of work of detachment in the edge and terrace direction can be neglected [22].

$$\Delta W_{K,i} = 2 \phi_{K,i} = 2 \gamma_{K,i} a_{P,i} h \quad (12)$$

$\gamma_{K,i}$  is the kink surface energy for a face  $i$ . Under solution growth, this surface energy is subject to change depending on the solvent used. This is because interfacial interactions between the crystal surface and solution medium directly influence the energetics of kink formation and its stability. To note the effect of solvent, bulk interface approximation of interfacial

energy model can be used [10]. According to the model:

$$\gamma_K = \gamma_X + \gamma_S - W_{ad,XS} \quad (13)$$

where  $\gamma_X$  is the surface energy of the pure crystal grown in vacuum;  $\gamma_S$  is the surface energy of pure solvent and  $W_{ad,XS}$  is the work of adhesion [23]. The model requires  $\gamma_X$  of every face  $i$  as the input, which can be found out experimentally or using crystal structure prediction software. The surface free energy of a pure solvent ( $\gamma_S$ ) is numerically equal to the surface tension of the pure solvent at specific temperature [24]. The work done by Girifalco and Good [25] provides a reliable approximation for calculating the work of adhesion. According to the theory, the work of adhesion can be given by:

$$W_{ad,XS} = 2\sqrt{\gamma_X \gamma_S} \quad (14)$$

The interfacial energy model, therefore, captures the effect of solvent on the surface energetics and can be used for different solvents at different temperatures.

As the supersaturation increases and there is sufficient driving force to overcome creation of new surface steps, crystal face starts growing through 2D nucleation mechanism. Small nuclei spread laterally along the crystal face to complete a layer. This is called Nucleation. In 2D Nucleation mechanism, it is the face coverage time used in Equation 3 [17], [26]. Within 2D Nucleation, two distinct sub regimes exist depending on the rate of nucleation,  $J$ . At very low  $J$ , spiral growth mechanism still dominates the crystal growth however as the  $J$  increases, multiple nuclei compete for layer completion. Based on this, the two distinct regimes are – Birth and spread Regime and Polynuclear Regime [17], [26]. In Birth and Spread regime, the initial nucleated area is low whereas in polynuclear regime, the growth area is low [27], [28]. The nucleation rate  $J$  is directly proportional to the supersaturation. Thus, as the supersaturation increases, the 2D nucleation mechanism becomes more dominant than the spiral growth mechanism. Within 2D nucleation, domination of Birth and Spread regime or Polynuclear regime also depends on supersaturation. It is essential to calculate the crossover supersaturation for the sub regimes within the 2D nucleation mechanism. To achieve this, the growth rates of the sub regimes can be modelled as functions of supersaturation [10]. Equating the growth rates and solving for the supersaturations allows us to obtain the crossover supersaturations.

The growth rate for 2D Birth and Spread mechanism can be written as a function of supersaturation as:

$$G_{2D,B+S}(i) = h_i (J_i f_i)^{\frac{1}{3}} \quad (15)$$

$$J_i = k_{2D,i} \exp\left(\frac{-3 F_i}{\ln(1 + \sigma)}\right) \quad (16)$$

$$k_{2D,i} = k_{2D,i}^* k_0 (1 + \sigma) \ln(1 + \sigma)^{\frac{1}{2}} \quad (17)$$

$$f_i = f_i^* \sigma^2 k_0^2 \quad (18)$$

$h_i, F_i, k_{2D,i}^*$  and  $f_i^*$  are face specific parameters which depend on the crystallography and are independent of the supersaturation, overall rate constant or temperature [10].

Like the 2D Birth and Spread mechanism, the growth rate for 2D Polynuclear mechanism is given by the expression:

$$G_{2D,P}(i) = h_i J_i A_{c,i} \quad (19)$$

$$A_{c,i} = \left( \frac{A_{c,i}^*}{(\ln(1 + \sigma))^2} \right) \quad (20)$$

Thus, equations 6, 15 and 19 provide expressions for calculating the growth rate for a crystal face for different mechanisms. However, depending on the supersaturation, the crystal will grow via either spiral mechanism, 2D Birth and Spread mechanism or 2D Polynuclear mechanism. The crossover supersaturation signifies the critical supersaturation level at which the dominant mechanism of a face will change from spiral to 2D Birth and Spread or from 2D Birth and Spread to Polynuclear and vice versa. At crossover supersaturations, growth rates of the two respective growth mechanisms will be equal [10]. The crossover supersaturations will be calculated for every face ( $i$ ) and will be dependent on the solvent used and the temperature of the system. To calculate the crossover supersaturation from spiral to 2D Birth and Spread ( $\sigma_{BS}(i)$ ), Equations 6 and 15 need to be equated by replacing  $\sigma$  term with  $\sigma_{BS}(i)$  term. Similarly, to obtain a numerical value of  $\sigma_P(i)$  (crossover supersaturation for 2D Birth and Spread to 2D Polynuclear),  $\sigma$  term is replaced with  $\sigma_P(i)$  term while equating Equations 15 and 19.

As the supersaturation increases, smaller nuclei are formed. In case of very small nuclei, the key assumption in the mechanistic model for defined edges with constant kink densities does not apply. Thus, the value of supersaturation where the nucleus size falls below 10 growth units can be considered as the upper limit for the applicability of the model [10]. The upper limit for supersaturation can be numerically obtained using the shape of factor of 2D nucleus ( $S_F$ ) and edge surface energy ( $\phi_E$ ),

$$\sigma_{NC} = \exp\left(\frac{1}{2\sqrt{10}} \frac{S_F \phi_E}{k_B T}\right) - 1 \quad (21)$$

As the supersaturation is increased beyond  $\sigma_{NC}$ , crystal surfaces become irregular. These irregular surfaces allow rapid and less selective growth of crystal faces. Further increase in supersaturation, forces the crystal faces to grow through bulk transport of solute molecules [29]. Both the growth mechanisms are responsible for non-uniform growth in crystals leading to irregular and undesirable crystal shapes [30], [31]. Therefore, for the current modelling framework, both these growth mechanisms are not considered. Figure 2

shows the dominant growth mechanisms at different supersaturation levels. It also illustrates the crossover supersaturations and the upper limit for supersaturation.

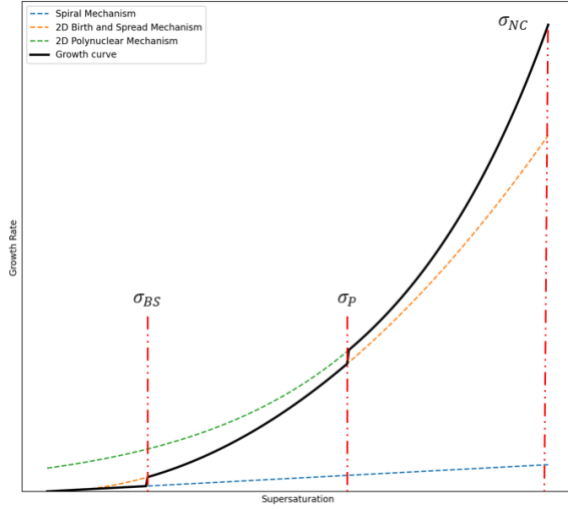


Figure 2 - Dominant Growth mechanisms, crossover supersaturations and supersaturation limit

The mechanistic model on its own cannot decide which mechanism of growth is applicable to the crystal face depending on the supersaturation. To facilitate the selection of growth mechanisms, binary variables namely  $z_S(i)$ ,  $z_{BS}(i)$  and  $z_P(i)$  are defined for spiral, 2D Birth and Spread and 2D Polynuclear mechanisms respectively. These variables denote the dominance of a growth mechanism on a crystal face. If a growth mechanism is dominant, the binary variable is assigned the value as 1, otherwise it is assigned with the value of 0. Generalised Disjunctive programming [32] is used to put constraints on the values of  $z_S(i)$ ,  $z_{BS}(i)$  and  $z_P(i)$  depending on supersaturation and crossover supersaturations for a face ( $i$ ). The constraints on the binary variables are listed in the Optimisation block. On any face  $i$ , only one of the growth mechanisms is dominant. This can be written mathematically as:

$$z_S(i) + z_{BS}(i) + z_P(i) = 1 \quad (22)$$

The growth rate of a face ( $i$ ) can be written as a function of these binary variables and the growth rate of the mechanisms. This allows selection of dominant growth mechanism for growth rate of the face ( $i$ ).

$$G_i = z_S(i) G_S(i) + z_{BS}(i) G_{2D,B+S}(i) + z_P(i) G_{2D,P}(i) \quad (23)$$

While doing crystal shape prediction, the supersaturation is already known and thus the values of the binary variables need to be assigned depending on the crossover supersaturations for the different faces. However, while doing the optimisation i.e. crystal shape control, the values of the binary variables will be selected by the modelling framework using the constraints given in optimisation block.

### Process Model

In cooling crystallisation processes, a solution containing dissolved solute and a polymorph seed is systematically cooled to create supersaturation [33]. As a result of supersaturation, crystal layers grow on the polymorph seed and give the resulting crystal a distinct shape. The cooling crystallisation system, in practice, follows the path given by crystal growth curve shown in Figure 3. However, directly simulating the system requires including a cooling strategy integrated with the mechanistic model. Rate of cooling in itself is an important factor influencing crystal shape [34], [35]. Therefore, for the current modelling framework, an arbitrary rate of cooling is assumed for final crystal shape prediction. To make the complex cooling crystallisation process computationally easy, we propose decomposing the continuous cooling process into discrete, sequential stages – Growth and Cooling stages (Figure 3). This simplification simulates the growth of crystals at distinct temperatures. The process starts from a supersaturated state (Point A). The vertical growth stages help the mechanistic model to compute supersaturation levels at specific temperatures. Specifying a cooling rate provides a next temperature point for the system. The new growth stage calculates the supersaturation at this new temperature point. The cooling stages connect the consecutive growth stages to show the decrease in temperature. Final shape of the crystal can be predicted by aggregating the shape prediction results at different temperature points. Therefore, in this study, shape predictions are made at constant temperatures to demonstrate the feasibility of the approach. The crystalliser Temperature is allowed to vary in the optimisation as this corresponds to the starting point temperature for the crystallisation process which provides maximum growth yield and minimum solvent consumption.

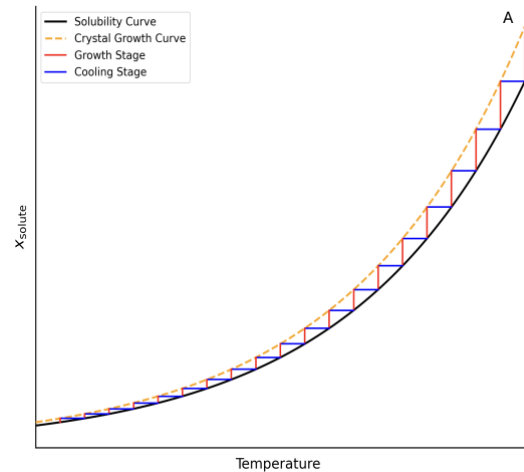


Figure 3 - Simplification of cooling crystallisation process

At constant temperatures, the yield of the crystal refers to the amount of solute that is deposited on the crystal face. This is termed as the growth yield for a specific temperature. The growth yield is given by:

$$Z = \frac{n_{solute} - n_{sat}}{n_{solute}} \quad (24)$$

$n_{solute}$  refers to the number of moles of solute at the start of the growth stage and  $n_{sat}$  is the number of moles of solute at the solubility limit. As it can be seen, the growth yield varies significantly with the solvent selected and the temperature at which the growth occurs.

Solvent consumption refers to the mass of solvent required per unit mass of solute to dissolve and subsequently crystallize the solute. It is typically expressed as grams of solvent per gram of solute. Solvent consumption is very important for designing the crystallisation processes. The solvent consumption also depends on the solvent that is chosen for the process and the temperature of the process. It is therefore used as a metric to optimise crystallisation processes for efficiency and product quality. The solvent consumption during the process of growth can be given by:

$$\chi_s = \frac{n_{sat}^{solvent}}{n_{initial}^{solute} - n_{sat}^{solute}} \frac{M.Wt.solute}{M.Wt.solvent} \quad (25)$$

The solubility for the solute is determined using the form:

$$\ln x_{sat}^{solute} \gamma_{sat}^{solute} = \frac{\Delta H_m^{solute}}{R} \left[ \frac{1}{T^m} - \frac{1}{T} \right] \quad (26)$$

$\gamma_{sat}^{solute}$  is the liquid phase activity coefficient of the solute at temperature  $T$  which is calculated using SAFT- $\gamma$  Mie.  $\Delta H_m^{solute}$  is the enthalpy of melting for solute which can be determined experimentally.  $T^m$  refers to the melting temperature for the solute,  $T$  is the temperature at which the crystal growth occurs, and  $R$  is the universal gas constant [36]. It is assumed that there is no loss of solvent during the process of crystallisation. Therefore, number of moles of solvent at the start of growth stage and the end of the growth stage are constant.

In cases where the process finds that the aspect ratio  $> 10$ , the system can suggest addition of a milling step to process design for shape control.

### **Optimisation Block**

The optimisation block serves as a critical component for controlling the crystal shape and meeting specific process objectives in crystallisation operations. The optimisation block integrates the mechanistic model and process model from the prediction block to systematically evaluate the effects of varying key process variables like choice of solvent, amount of chosen solvent, and temperature on crystal shape, growth yield, and solvent consumption. The optimisation framework incorporates an additional constraint set that enables selective activation of dominant growth mechanisms for specific crystal faces. The objective function, decision variables and process constraints need to be specified to obtain a

reliable and optimal solution using the optimisation block.

### ***Objective Function***

The optimisation framework accommodates both single-objective and multi-objective paradigms to address complex trade-offs in industrial crystallisation. From a chemical engineering perspective, yield maximisation at constant temperature and starting conditions ensures maximum amount of solute is converted to a crystal, optimising process efficiency. When sustainability is prioritised, solvent consumption minimisation can become the objective function, reducing environmental impact and operational costs. For applications requiring block shaped crystals, aspect ratio minimization ensures cubic morphologies essential for specific downstream processing requirements. These objectives can be effectively combined using multi-objective approaches ( $\min F = \sum w_i f_i$ ) where weights ( $w_i$ ) reflect process priorities and economic considerations. For the current modelling framework, the objective function for the optimisation block is maximisation of growth yield and minimisation of solvent consumption. This can be expressed mathematically as :

$$\max_{n,solvent,T} Z - \chi_s \quad (27)$$

### ***Decision Variables***

The objective of the optimisation block is to control the crystal shape by varying the growth conditions of the crystal. Thus, the decision variables include the variables which are responsible for change in growth conditions of the crystal. The primary decision variable for the growth process is the solvent used for growth of crystal. Different solvents have different solubility limits and different surface free energies which alter the growth shape of the crystals. The amount of selected solvent added is another decision variable. The amount of solvent added remains constant throughout the process because there is no solvent loss during the process of crystallisation. The amount of solvent added helps to change the supersaturation level which is responsible for dictating the growth mechanism for different crystal faces. Temperature constitutes the last decision variable for the optimisation process. To simplify the design space, temperature can be omitted as the decision variable, and the growth process can be carried out a fixed given temperature. The temperature has an influence on the solubility of solute in the solvent and thus changes the supersaturation level of the solution.

### ***Binary Variable Constraints***

Generalised Disjunctive programming [32] is used to impose constraints on the binary variables  $z_S(i)$ ,  $z_{BS}(i)$  and  $z_P(i)$  to allow selection of the dominant growth mechanism. Below  $\sigma_{BS}(i)$ , spiral growth mechanism is dominant implying  $z_S(i) = 1$ . When supersaturation level lies between  $\sigma_{BS}(i)$  and  $\sigma_P(i)$ , 2D Birth and Spread mechanism is dominant which in turn implies that  $z_{BS}(i) = 1$ . After the

supersaturation level crosses  $\sigma_p(i)$  threshold, 2D Polynuclear growth mechanism becomes dominant ( $z_p(i) = 1$ ). Equations 28 – 31 are included in the modelling framework to allow assignment of binary variables depending on the supersaturation level.

$$\sigma - \sigma_{BS}(i) - M(1 - z_S(i)) \leq 0 \quad (28)$$

$$\sigma - \sigma_{BS}(i) + M(1 - z_{BS}(i)) \geq 0 \quad (29)$$

$$\sigma - \sigma_p(i) - M(1 - z_{BS}(i)) \leq 0 \quad (30)$$

$$\sigma - \sigma_p(i) + M(1 - z_p(i)) \geq 0 \quad (31)$$

$\sigma_{BS}(i)$  and  $\sigma_p(i)$  denote the crossover supersaturations for a crystal face  $i$  for the selected solvent.  $M$  is a constant within the range  $M \in (0, 0.1)$ . This approach chooses the values of binary variables according to the supersaturation and allows to calculate the growth rate of a crystal face.

#### Process Constraints

Logical constraints are used to ensure that only one solvent is selected out of all the solvent candidates ( $N_S$ ) for the process.

$$\sum_{i \in N_S} y_i = 1 \quad (32)$$

The optimisation block must control the shape of crystals to avoid needle shaped crystals. It has been observed that that crystals with aspect ratio  $> 10$  exhibit needle like shape which is highly unfavourable. The aspect ratio of 1 signifies that all the crystal faces grow at a uniform rate which in turn gives the crystal a block like shape. Therefore, the constraints imposed on aspect ratio are:

$$1 \leq \text{Aspect Ratio} \leq 10 \quad (33)$$

The mechanistic model fails when very small nuclei are formed because of high supersaturation. Therefore, there needs to be an upper limit on the supersaturation of the solution which is given by Equation 21.

Additional constraints can be added depending on the optimisation problem that is selected. There may be additional constraints on the solvent consumption or growth yield to ensure that the process is either sustainable or efficient. Depending on the heuristics, constraints on solvent consumption can help to maintain the feasibility and operability of the process.

Though some optimisation results may seem mathematically relevant and optimal, it is necessary to ensure that the process remains feasible. It is necessary to ensure that the selected solvent does not freeze or boil during the process of crystallisation. Thus,

$$T_{min} \leq T \leq T_{max} \quad (34)$$

where  $T_{min}$  is the highest freezing point of the solvent among the solvent candidates,  $T$  is the crystalliser temperature and  $T_{max}$  is the lowest boiling point of the solvent among the solvent candidates.

## Results and Discussion

The proposed modelling framework of crystal shape prediction and control is implemented in gPROMS. gSAFT within gPROMS is utilised for thermodynamic and solubility calculations based on SAFT- $\gamma$  Mie Equation of State (EoS) [37], [38]. Two case studies - crystallisation of Biphenyl and Naphthalene are considered to validate the mechanistic model, predict shape of crystals in different growth conditions and optimise the respective crystallisation processes for desirable crystal growth. The crystalliser temperature is kept constant for the prediction block to validate the model results with the experimental results and facilitate easy comparison between growth shapes predicted in different solvents. In the optimisation block, the crystalliser temperature is allowed to vary. This temperature corresponds to the starting point for the crystallisation process. The optimisation block provides an optimal solvent for carrying out the crystallisation process, optimal amount of solvent to be added during the start of crystallisation process and the optimal starting Temperature for the crystallisation process. On obtaining the optimal values of decision variables, the process is simulated by feeding an arbitrary cooling strategy to estimate the final shape of crystal and estimate the process yield.

### Case Study 1 – Crystallisation of Biphenyl

Biphenyl is an important intermediate in various industrial processes and serves as a precursor for pharmaceutical compounds, agrochemicals and speciality chemicals. Conventionally, Biphenyl crystals are grown using Toluene as a solvent [39]. However, this produces plate like biphenyl crystals which are difficult to wash, filter and handle in an industrial setting. This leads to low product yield and bottlenecks which provides an opportunity for the crystal shape prediction and control framework to provide optimal growth conditions for control of crystal shape

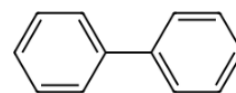


Figure 4 - Chemical Structure of Biphenyl

Biphenyl is an aromatic hydrocarbon consisting of 2 phenyl rings connected by a C-C bond. Figure 4 represents the structure of the biphenyl molecule. SAFT- $\gamma$  Mie group contribution method has been shown to predict the solubility of Biphenyl in different candidate solvents over different temperatures accurately [40]. The mechanistic parameters and crystallographic inputs required for the modelling framework were obtained from the works of Tilbury and Doherty [10] and Winn and Doherty [41]

(Supplementary Information). The same mechanistic parameters can also be obtained using the Cambridge Structural Database (CSD) CIF file BIPHEN04 [42].

Table 1 - Crystal shape predictions for Biphenyl grown from Toluene

|                     | Face  |        |        |
|---------------------|-------|--------|--------|
|                     | {001} | {110}  | {111}  |
| $R(\sigma = 0.005)$ | 1     | 1.3    | 16.8   |
| $R(\sigma = 0.006)$ | 1     | 3.1    | 85     |
| $R(\sigma = 0.007)$ | 1     | 18     | 252    |
| $\sigma_{BS}$       | -     | 0.0053 | 0.0029 |
| $\sigma_P$          | -     | 0.0061 | 0.0038 |
| $\sigma_{NC}$       | 0.93  | 0.1    | 0.09   |

The mechanistic model was validated by considering growth of Biphenyl in Toluene at  $T = 302$  K. Growth of Biphenyl in toluene is well reported and studied experimentally [43]. Prediction studies have also been performed previously using mechanistic models for spiral and 2D nucleation mechanism. Table 1 summarizes the predicted relative growth rates (with respect to face {001}) and the crossover supersaturations for the crystal faces. The predicted relative growth rates at  $\sigma = 0.005$  and  $\sigma = 0.006$  match exactly with the experimentally reported results and previous prediction studies. The predicted relative growth rate at  $\sigma = 0.007$  varies from the experimentally reported previous simulation results by  $\sim 10\%$ . The source of the error can be attributed to the approximation of the overall rate constant and minor variations in crystallographic parameters with temperature.

The model also calculates the crossover supersaturations for the crystal faces. It is observed that the {111} face family crosses over 2D polynuclear mechanism at very low supersaturation of 0.004 and eventually grow out of the crystal shape. The {110}

face family has a predicted crossover to 2D Birth and Spread at supersaturation of 0.0053 and the face crosses over to 2D Polynuclear mechanism at  $\sigma = 0.0061$ . The face family {001} does not crossover to 2D Birth and Spread or 2D Polynuclear mechanism because of very high crossover supersaturations (beyond the upper bound constraint on supersaturation i.e.  $\sigma_{BS \text{ or } P} \geq \sigma_{NC}$ ). It is the crossover supersaturation, that determines the dominant growth mechanism and which in turn changes the shape of crystal. Figure 5 shows the shapes of crystal and its changes in aspect ratio at  $\sigma = 0.005$  and  $\sigma = 0.006$ . The aspect ratio of the crystal changes from 1.3 to 3.1. The model predicts a rhombic platelet shape for biphenyl grown from toluene which agrees with the experimentally reported data [43].

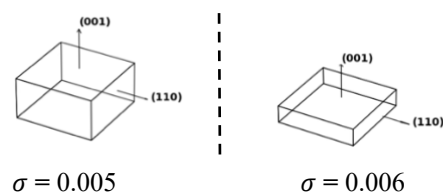


Figure 5 - Change in crystal shape of biphenyl with supersaturation

To note the effect of solvent on the crystal shape, growth yield and solvent consumption, predictions were made for the same supersaturation levels i.e.  $\sigma = 0.005$ ,  $\sigma = 0.006$  and  $\sigma = 0.007$  at Temperature  $T = 302$  K using different solvents. A total of 10 candidate solvents (Table 3) were used to predict the shape of crystals and provide insights regarding yield and solvent consumption. The variations in aspect ratio, growth yield, and solvent consumption demonstrate the influence of solvent on the crystallisation of naphthalene (Table 2). At low supersaturation ( $\sigma = 0.005$ ), spiral growth predominates, and toluene

Table 2 - Effect of solvent on Aspect Ratio, growth yield and solvent consumption for Biphenyl Crystallisation process

| Solvent       | $\sigma = 0.005$ |       |                | $\sigma = 0.006$ |       |                | $\sigma = 0.007$ |       |                |
|---------------|------------------|-------|----------------|------------------|-------|----------------|------------------|-------|----------------|
|               | Aspect Ratio     | Z (%) | $\chi_s$ (g/g) | Aspect Ratio     | Z (%) | $\chi_s$ (g/g) | Aspect Ratio     | Z (%) | $\chi_s$ (g/g) |
| Methanol      | 1.21             | 0.50  | 4143.48        | 3.42             | 0.60  | 3416.84        | 19.89            | 0.70  | 2959.08        |
| Ethanol       | 1.22             | 0.50  | 5640.14        | 3.39             | 0.60  | 4700.42        | 19.72            | 0.70  | 4026.97        |
| Acetic Acid   | 1.32             | 0.50  | 13416.3        | 3.12             | 0.59  | 11180.8        | 18.11            | 0.69  | 9581.80        |
| Butanol       | 1.27             | 0.50  | 9345.56        | 3.25             | 0.95  | 4887.11        | 18.90            | 0.70  | 6620.09        |
| Butyl Acetate | 1.28             | 0.57  | 870.97         | 3.22             | 0.68  | 725.68         | 18.71            | 0.79  | 621.90         |
| Diethyl Ether | 1.05             | 0.51  | 3291.19        | 3.91             | 0.61  | 2909.2         | 6.52             | 0.71  | 2493.48        |
| Acetone       | 1.25             | 0.50  | 4508.69        | 3.30             | 0.60  | 3755.69        | 19.21            | 0.71  | 3163.41        |
| Hexane        | 1.10             | 0.52  | 1991.73        | 3.75             | 0.62  | 1659.68        | 6.25             | 0.73  | 1422.51        |
| Cyclohexane   | 1.28             | 0.50  | 10739.5        | 3.22             | 0.60  | 8955.01        | 18.69            | 0.71  | 7554.61        |
| Toluene       | 1.33             | 0.69  | 218.08         | 3.10             | 0.83  | 181.66         | 18               | 0.96  | 155.65         |

achieves the highest yield with minimal solvent usage. Toluene facilitates efficient crystal growth with reduced solvent requirements; however, the resulting crystals are less compact compared to those produced with diethyl ether. As supersaturation increases to  $\sigma = 0.006$ , the dominant growth mechanism shifts, resulting in diethyl ether producing crystals with the highest aspect ratios, while toluene yields the most compact crystals (aspect ratio 3.1). This transition indicates that alterations in growth mechanisms can reverse the influence of solvents on crystal morphology. Toluene consistently requires the least solvent, likely due to its high solubility for naphthalene. At  $\sigma = 0.006$ , butanol provides the highest yield, indicating that optimal solvent selection depends on both supersaturation and growth conditions. Acetic acid consistently exhibits the lowest yield and highest solvent consumption, which may be attributed to its poor interaction with the crystal surface.

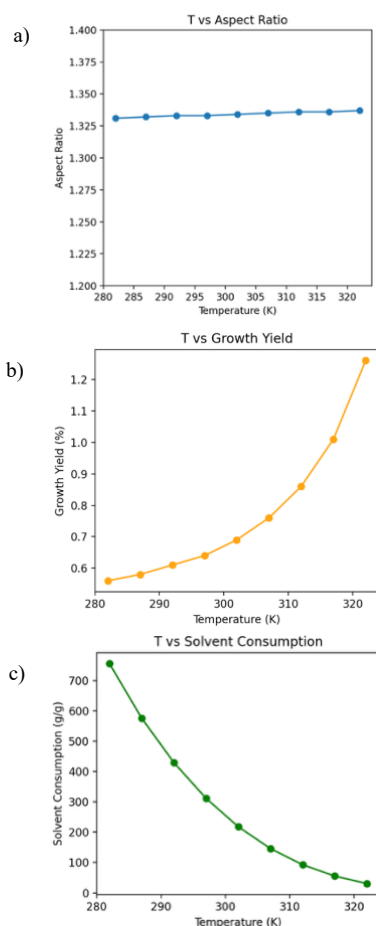


Figure 6 - Variation in aspect ratio, growth yield and solvent consumption with respect to temperature for Biphenyl Crystallisation

The effect of temperature on the growth shape, growth yield and solvent consumption was also observed. For effective comparison, toluene was selected as the solvent, and the supersaturation was kept fixed at 0.005. The crystalliser temperature was varied from 282 K to 322 K. Figure 6 a,b and c represent the variation in aspect ratio, growth yield and solvent consumption. The variation in aspect ratio is

minimum. This is because the surface energy of solvent and pure crystal decrease with increase in temperature, effectively offsetting the effect of temperature change. The variations in growth yield and solvent consumption are due to the changes in solubility limit of biphenyl in toluene at different temperatures. High yield and low solvent consumption is possible at high temperatures while as the temperature is reduced the solubility of biphenyl also reduces giving low growth yield and high solvent consumption. It is also observed that as the temperature is changed, the crossover supersaturations for the crystal faces also change.

Table 3 - Inputs to optimisation block, candidate solvents and process model limits for Biphenyl Crystallisation

| Description                   | Inputs                                                                                                       |
|-------------------------------|--------------------------------------------------------------------------------------------------------------|
| <i>Molecule</i>               | Biphenyl                                                                                                     |
| <i>Number of Solvents</i>     | 10                                                                                                           |
| <i>Candidate Solvents</i>     | Methanol, Ethanol, Acetic Acid, Butanol, Butyl Acetate, Diethyl Ether, Acetone, Toluene, Hexane, Cyclohexane |
| <i>Temperature limits</i>     | $T_{min} = 289\text{ K}$ and $T_{max} = 308\text{ K}$                                                        |
| <i>Supersaturation limits</i> | $\sigma_{min} = 0.004$ and $\sigma_{max} = 0.09$                                                             |
| <i>Aspect Ratio limits</i>    | $0 \leq \text{Aspect Ratio} \leq 10$                                                                         |

The main objective of the optimisation problem is to maximise the growth yield while minimising the solvent consumption. The aspect ratio of the crystals is bounded by an upper bound of 10 to avoid needle shaped crystals. As a result, through the optimiser, we obtain the optimal solvent and the corresponding process conditions i.e. amount (moles) of solvent required and the starting temperature of crystalliser. The candidate solvents used in the optimisation problem are provided in Table 3. The bounds on the decision variables and process variables are also provided. Table 4 ranks the top optimal solvent choices for maximising the growth yield and minimising the solvent consumption. For maximising the yield and minimising the solvent consumption it is noted that crystallisation needs started at high temperatures. Through the optimiser, the optimal value of starting temperature is found to be 308 K. Toluene is the best choice of solvent for maximising the growth yield and minimising solvent consumption. However, we need to have a trade off with respect to aspect ratio as high yield comes at the cost of aspect ratio. The aspect ratio of crystals is almost equal to 10 giving them a flatter needle like shape. To control this and eliminate completely, the limits on aspect ratio can be further restricted. The growth yield for crystals grown in Toluene is 1.02 % and the solvent consumption is

Table 4 - Optimisation results for Biphenyl Crystallisation process

| Rank | T (K) | Solvent       | Aspect Ratio | Z (%) | $\chi_s$ (g/g) |
|------|-------|---------------|--------------|-------|----------------|
| 1    | 308   | Toluene       | 9.97         | 1.02  | 101.33         |
| 2    |       | Butyl Acetate | 9.8          | 0.78  | 451.04         |
| 3    |       | Hexane        | 5.18         | 0.7   | 1124.6         |

101.33 g/g. The second-best solvent for achieving the set objective is Butyl Acetate. The resulting crystals have an aspect ratio of 9.8 and a growth yield of 0.78%. The solvent consumption is 451.04 g/g which is higher than that of Toluene. The third best solvent choice for crystallisation of biphenyl is Hexane. Because Face {110} grows by 2D Birth and Spread mechanism, the aspect ratio of crystals grown in Toluene is 5.18. The growth yield is 0.7 % and 1124.6 g/g of solvent is consumed to produce 1 g of crystal.

The optimisation block provides the optimal solvent, optimal process conditions. However, because cooling crystallisation is not a one step process, it is important to test the model by providing a cooling strategy to understand the final shape of crystal at the end of crystallisation process. For the current model, the rate of cooling is kept constant at 0.1K / s. The system starts from 308 K and is cooled instantaneously after 1 second of growth. This effectively reduces the temperature of the system by 0.1 K inducing supersaturation in the system. The system is maintained at this temperature for another 1 second to facilitate growth of crystals. The process is run for 10 seconds to simulate the progression of cooling crystallisation process and observe the evolution of crystal shape. The cooling curve and variation in aspect ratio with temperature is shown in Figure 7 a,b – In the first 1 second, the aspect ratio increases signifying the growth of crystal from polymorph seed as a result of supersaturation  $\sigma = 0.00653$ . As the

temperature is reduced steadily at a rate of 0.1K /, the aspect ratio is reduced from 9.86 to 2.61. The final process yield of 7.34 % is obtained when the system is cooled uniformly from 308 K to 307 K.

### Case Study 2 – Crystallisation of Naphthalene

Naphthalene is a polycyclic aromatic hydrocarbon which has applications from production of bulk chemicals to speciality chemicals like pharmaceuticals, dyes and resins. The morphology of naphthalene crystals is observed to vary significantly between hexagonal plates, needles, dendrites, and some other irregular shapes [44]. Using crystal shape prediction and control framework, we can avoid the formation of undesirable crystal shapes to improve efficiency of the crystallisation process.

Figure 8 shows the chemical structure of Naphthalene molecule. SAFT- $\gamma$  Mie group contribution method is used predict solubility of Naphthalene in different candidate solvents over different temperatures [40]. The mechanistic parameters required for the model are obtained from the works of Tilbury and Doherty [10]. The values of the mechanistic parameters used for the study are provided in the Supplementary Information. The same set of mechanistic parameters can be obtained using Cambridge Structural Database (CSD) CIF code file NAPHTA10 [45].

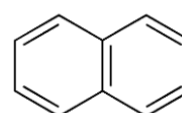


Figure 8 - Chemical Structure of Naphthalene

To validate the mechanistic model, crystal shape predictions were made for naphthalene grown from ethanol at 291 K which is experimentally a well reported case [44]. Table 5 summarises the relative growth rates of naphthalene faces at different supersaturations at 291 K. The relative growth rates can be used further to calculate the aspect ratio and shape of crystal. The prediction model also calculates the crossover supersaturations for the crystal faces. It is noted that face family {001} does not crossover to 2D nucleation mechanism which was validated by the model. Face Families {201} and {110} crossover from Spiral growth mechanism to 2D Birth and Spread mechanism at  $\sigma = 0.36$  and  $\sigma = 0.29$  respectively. No crossover to 2D polynuclear mechanism is predicted for these two faces. The crossover supersaturations for

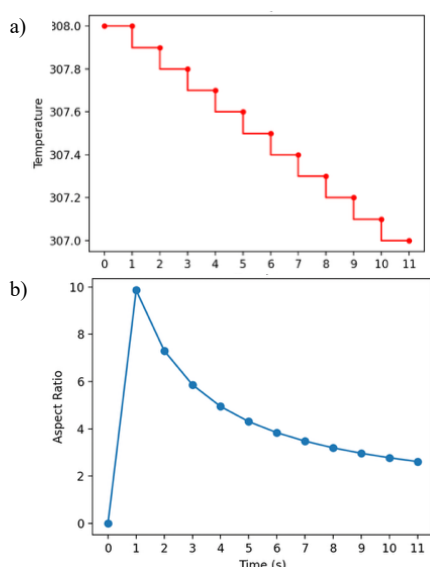


Figure 7 a,b – Cooling curve and variation in aspect ratio for Biphenyl Crystallisation

face  $\{11\bar{1}\}$  are  $\sigma_{BS} = 0.13$  and  $\sigma_P = 0.35$ . The predictions of relative growth rate (shape) at  $\sigma < 0.13$  and  $\sigma = 0.16$ , align perfectly with the predictions made by Tilbury and Doherty using the mechanistic model [10]. The relative growth rate (shape) predictions vary from the results obtained in previous study by a margin of  $\sim 0.1 - 3\%$ .

Table 5 - Crystal Shape prediction for Naphthalene grown in Ethanol

|                     | Face      |                 |           |                 |
|---------------------|-----------|-----------------|-----------|-----------------|
|                     | $\{001\}$ | $\{20\bar{1}\}$ | $\{110\}$ | $\{11\bar{1}\}$ |
| $R (\sigma < 0.13)$ | 1         | 1.23            | 1.18      | 1.128           |
| $R (\sigma = 0.16)$ | 1         | 1.23            | 1.18      | 2.3             |
| $R (\sigma = 0.35)$ | 1         | 1.24            | 1.56      | 5.77            |
| $\sigma_{BS}$       | -         | 0.36            | 0.29      | 0.13            |
| $\sigma_P$          | -         | -               | -         | 0.35            |
| $\sigma_{NC}$       | 3.1       | 0.7             | 0.67      | 0.48            |

Solvent choice greatly affects the crystal shape and the process conditions. To study this effect, the growth shape of naphthalene when grown from different solvents at supersaturation levels of  $\sigma = 0.1$  and  $\sigma = 0.48$  at  $T = 291$  K was studied. A total of 10 commonly used industrial solvents are used as candidates for predicting the crystal shape and process parameters. The candidate solvents are listed in Table 7. However, at 291 K, the solubility of naphthalene in solvents like Water, Butanol, Pentanol and Diethyl Ether is very low. This makes crystallisation of naphthalene at low temperatures in these solvents very difficult. Therefore, the results for shape predictions for these 4 solvents are not reported. Shape predictions for crystals grown from the rest of the solvents are listed in Table 6. At higher supersaturations of  $\sigma = 0.48$ , crystal grown from DMSO exhibit highest aspect ratio while showing lowest growth yield and highest solvent consumption. Crystals grown from Ethanol have lowest aspect ratio of 8.00 among other solvents. At  $\sigma = 0.1$ , for all solvents the aspect ratio lies in the range 1.2-1.4. Highest crystal growth yield is reported when

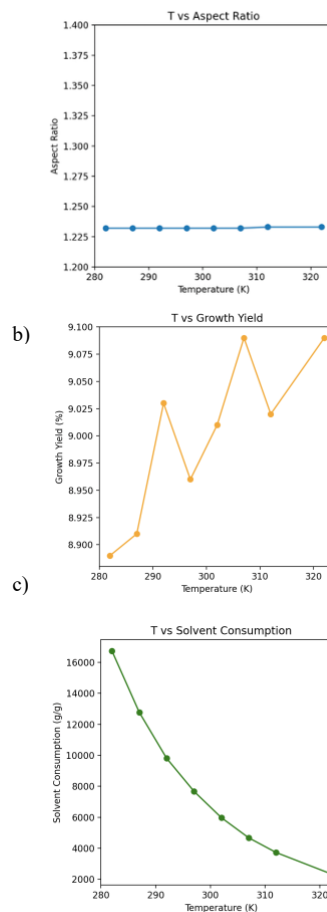


Figure 9 - Variation in aspect ratio, growth yield and solvent consumption with respect to temperature for Naphthalene Crystallisation.

Methanol is used as a solvent while for 1-Pentanol as the solvent, lowest solvent consumption is observed. DMSO produces crystals with aspect ratio 1.39 and has the lowest growth yield and highest solvent consumption relative to the other candidate solvents. It can be seen from these results that high aspect ratios result from preferential growth along specific crystallographic directions due to strong solvent-surface interactions. Solvents with strong solvation of naphthalene (high solubility) require larger volumes to

Table 6 - Effect of solvent on aspect ratio, growth yield and solvent consumption for Naphthalene Crystallisation

| Solvent     | $\sigma = 0.1$ |       |                | $\sigma = 0.48$ |       |                |
|-------------|----------------|-------|----------------|-----------------|-------|----------------|
|             | Aspect Ratio   | Z (%) | $\chi_s$ (g/g) | Aspect Ratio    | Z (%) | $\chi_s$ (g/g) |
| Methanol    | 1.23           | 11.0  | 14528.5        | 8.00            | 32.41 | 3753.75        |
| Ethanol     | 1.23           | 9.08  | 10274.5        | 8.00            | 32.3  | 2140.38        |
| Propanol    | 1.22           | 8.96  | 8465.48        | 8.01            | 32.41 | 1744.51        |
| 1-Pentanol  | 1.22           | 9.04  | 7429.81        | -               | -     | -              |
| Acetic Acid | -              | -     | -              | 8.12            | 32.47 | 3025.22        |
| DMSO        | 1.39           | 8.83  | 23331.3        | 9.91            | 32.18 | 4793.37        |
| Acetone     | 1.22           | 9.11  | 1422.93        | 8.01            | 32.51 | 295.93         |

Table 7 - Inputs to Optimisation block, candidate solvents and process model limits for Naphthalene Crystallisation

| Description            | Inputs                                                                                            |
|------------------------|---------------------------------------------------------------------------------------------------|
| Molecule               | Naphthalene                                                                                       |
| Number of Solvents     | 10                                                                                                |
| Candidate Solvents     | Water, Methanol, Ethanol, Propanol, Butanol, Pentanol, Acetic Acid, Acetone, DMSO, Diethyl Ether. |
| Temperature limits     | $T_{min} = 285 \text{ K}$ and $T_{max} = 306 \text{ K}$                                           |
| Supersaturation limits | $\sigma_{min} = 0.1$ and $\sigma_{max} = 0.48$                                                    |

achieve target supersaturation levels, increasing consumption. Conversely, poor solvents like acetone require less volume but may limit maximum achievable supersaturation. At low supersaturation ( $\sigma = 0.1$ ), growth is thermodynamically controlled, minimizing solvent-dependent shape differences (aspect ratios 1.2-1.4). At high supersaturation ( $\sigma = 0.48$ ), kinetic effects dominate, amplifying solvent-specific surface interactions and creating larger aspect ratio variations.

To note the effect of temperature on crystal shape, growth yield and solvent consumption, a case of naphthalene crystallisation in Ethanol was considered. The supersaturation was fixed at  $\sigma = 0.1$  and the temperature was varied from 322 K to 282 K. Figure 9 a, b and c shows the variation in aspect ratio, growth yield and solvent consumption with respect to temperature.

The objective function for the process of crystallisation of naphthalene is to maximise the growth yield while minimising the solvent consumption. To avoid formation of needle shaped crystals, aspect ratio is bounded at 10. By solving the optimisation problem, an optimal solvent choice is obtained. The solution also suggests optimal process conditions to achieve the process objective. The candidate solvents used for the optimisation of naphthalene crystallisation process are listed in Table 7. The limits on process conditions and decision variables are also listed here.

Using the inputs provided to the optimisation block, the optimiser maximises the crystal growth

yield and minimises the solvent consumption for the growth stage. The results of the optimisation block are provided in Table 8. For maximising the yield and minimising the solvent consumption it is noted that crystallisation needs to be started at high temperatures. Through the optimiser, the optimal value of starting temperature is found to be 306 K. Acetone is found to be the best choice of solvent for maximising the growth yield and minimising solvent consumption. Naphthalene when grown in Acetone produces crystals with aspect ratio 8.01. A high growth yield of 32.62 % is reported and the solvent consumption is also lowest at 156.05 g/g. The second-best solvent for achieving the set objective is 1-Pentanol. The resulting crystals have an aspect ratio of 8.02 and a growth yield of 32.49 % which is almost comparable to the crystal shape and yield for Acetone. But the solvent consumption is 762.31 g/g which is higher than that of Acetone. The third solvent choice for crystallisation of naphthalene is Butanol. Aspect Ratio of the crystals obtained is 8.03 and the growth yield and solvent consumption is 32.48 % and 785.77 g/g respectively.

The results obtained through the optimisation block were simulated using the cooling crystallisation process model. Acetone was taken as the solvent and the starting point temperature for the crystalliser was fixed at 306 K. To provide a proof of concept for the model, an arbitrary cooling rate is specified. The temperature of the system is maintained constant for 1 second to facilitate growth. At the next time step, the temperature is reduced by 3 K. The process was simulated for two temperature reduction steps till the temperature of the crystalliser reaches 300 K. Beyond this point, it was observed that supersaturation was too low to drive the crystallisation process and very low crystal yields were obtained. The final process yield for the process was found to be 63.6 %. Figure 10 a, b shows the temperature change (cooling curve) and change in aspect ratio (shape). The aspect ratio of the naphthalene crystals reduces from 8.01 at the start of the process to 7.57 by the end of the process. This change is caused by variations in supersaturation induced due to cooling. Changing the cooling rate may significantly affect the final shape of crystals. Reducing the temperature by 1 K instead of 3 K may aid in promoting controlled crystal growth over longer growth times giving the crystal a block-like shape.

Table 8 - Optimisation results for Naphthalene crystallisation process

| Rank | T (K) | Solvent    | Aspect Ratio | Z (%) | $\chi_s$ (g/g) |
|------|-------|------------|--------------|-------|----------------|
| 1    | 306   | Acetone    | 8.01         | 32.62 | 156.05         |
| 2    |       | 1-Pentanol | 8.02         | 32.49 | 762.31         |
| 3    |       | Butanol    | 8.03         | 32.48 | 785.77         |

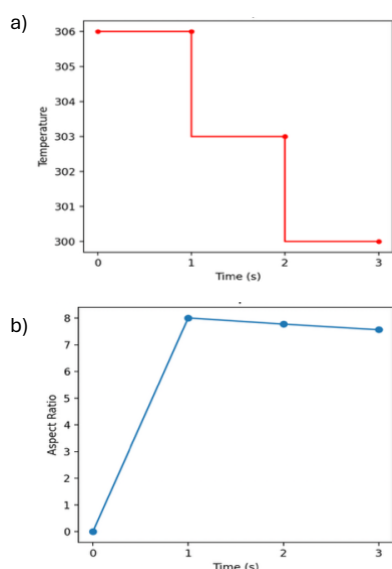


Figure 10 - Cooling curve and variation in aspect ratio for Naphthalene Crystallisation

## Conclusions

A comprehensive modelling framework for predicting and controlling the shape of crystals through solvent selection has been presented. The modelling framework aims to guide the experiments and provide a strategy for control of shape of crystals. Optimal solvent to be selected, amount of solvent to be added and the crystalliser temperature required to maximise the crystallisation yield while minimising solvent consumption can be identified.

The modelling framework is successfully applied to crystallisation of biphenyl where, the effect of change in solvent and temperature on crystal shape is shown. We show that selecting toluene as the solvent and starting the cooling crystallisation process from 308 K maximises the growth yield of the starting growth stage while reducing the solvent consumption. Furthermore, capability of the framework to predict the process yield and the final growth shape of the crystal (depending on the cooling strategy) is also demonstrated. Similarly, the framework is applied to the crystallisation process of naphthalene showing its robustness, versatility and general approach. Overall, the modelling framework is effective to predict and simultaneously control the growth shape of crystals while providing process level insights.

Simplification of the cooling crystallisation process model makes the prediction of crystal shape at specific temperatures computationally cheaper. In both the studies, an arbitrary cooling strategy is provided to test the prediction capability of the cooling crystallisation process model. It has been observed that rate of cooling is an important factor controlling the shape of crystals through the case studies.

The presented modelling framework is a proof-of-concept stage for demonstrating the possibility of development of a framework capable of controlling the shape of crystal while achieving the process objectives. Further improvements need to be made to advance this

type of study. The process model and the mechanistic model used in this modelling framework are simplifications of the actual process system built by considering several critical assumptions. This causes the results to vary over a margin of error. Better and improved mechanistic and process models can be used in the modelling framework to improve the accuracy of the results. Whilst solvent selection in the current model purely achieves the given objectives, toxicity of solvents, HSE considerations like flammability, reactivity may be used to screen the solvents in the future. In the further improvements of the model, the cooling strategy can be added as a key decision variable along with solvent selection to better optimise the cooling crystallisation process. Finally, it can be advantageous for the modelling framework to include wider process impacts like capital and operating costs associated with solvent-based cooling crystallisation processes.

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## □ Supplementary Information

The mechanistic model parameters for Biphenyl and Naphthalene are adopted from the work of Tilbury and Doherty [10]. Tables 9 and 10 list the mechanistic parameters for Faces {001} and {110} of Biphenyl crystal grown from Toluene.

Table 9 - Mechanistic parameters for Biphenyl Face {001} grown from Toluene [10]

| Face (001)          |                 |                  |          |
|---------------------|-----------------|------------------|----------|
| Parameter           |                 | Units            | Value    |
|                     | $h$             | nm               | 0.947    |
|                     | $\tau_S^*$      | -                | 5.68     |
|                     | $\kappa_{2D}^*$ | cm <sup>-2</sup> | 1.15e+14 |
|                     | $f^*$           | cm <sup>2</sup>  | 2.52e-16 |
|                     | $A_C^*$         | cm <sup>2</sup>  | 8.70e-15 |
|                     | $C_0^{2D}$      | cm <sup>-2</sup> | 3.74e+14 |
|                     | $s_F$           | -                | 3.74     |
|                     | $\mathcal{F}$   | -                | 1.45     |
| [010] PBC           | $H_C^{2D*}$     | nm               | 0.524    |
|                     | $\phi^K$        | $kT$             | 0.065    |
| [1 $\bar{1}$ 0] PBC | $H_C^{2D*}$     | nm               | 0.490    |
|                     | $\phi^K$        | $kT$             | 0.204    |
| [110] PBC           | $H_C^{2D*}$     | nm               | 0.490    |
|                     | $\phi^K$        | $kT$             | 0.204    |

Table 10 - Mechanistic parameters for Biphenyl Face {110} grown from Toluene [10]

| Face (110)          |                 |                  |          |
|---------------------|-----------------|------------------|----------|
| Parameter           |                 | Units            | Value    |
|                     | $h$             | nm               | 0.462    |
|                     | $\tau_S^*$      | -                | 3.82     |
|                     | $\kappa_{2D}^*$ | cm <sup>-2</sup> | 8.39e+13 |
|                     | $f^*$           | cm <sup>2</sup>  | 5.19e-16 |
|                     | $A_C^*$         | cm <sup>2</sup>  | 5.82e-16 |
|                     | $C_0^{2D}$      | cm <sup>-2</sup> | 1.46e+14 |
|                     | $s_F$           | -                | 4.36     |
|                     | $\mathcal{F}$   | -                | 0.029    |
| [001] PBC           | $H_C^{2D*}$     | nm               | 0.191    |
|                     | $\phi^K$        | $kT$             | 0.021    |
| [1 $\bar{1}$ 2] PBC | $H_C^{2D*}$     | nm               | 0.200    |
|                     | $\phi^K$        | $kT$             | 0.093    |
| [1 $\bar{1}$ 0] PBC | $H_C^{2D*}$     | nm               | 0.076    |
|                     | $\phi^K$        | $kT$             | 0.204    |

The solubility of Biphenyl in different solvents in the temperature range 10 – 35 °C is given in Figure 11

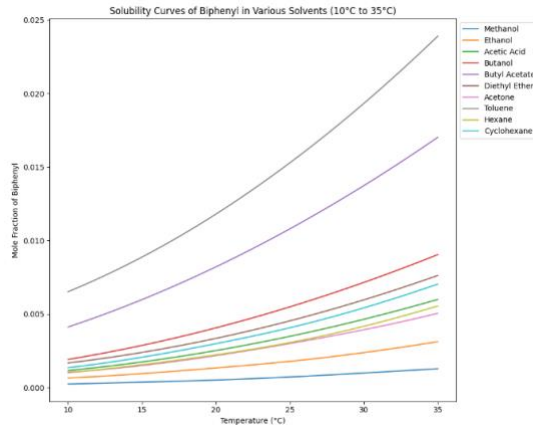


Figure 11 - Solubility of Biphenyl in different solvents

The mechanistic model parameters for faces {001}, {20 $\bar{1}$ }, {110} and {11 $\bar{1}$ } of Naphthalene grown in Ethanol are listed in Tables 11-14.

Table 12 - Mechanistic parameters for Naphthalene Face {001} grown from Ethanol [10]

| Face (001)      |                  |          |      |
|-----------------|------------------|----------|------|
| Parameter       | Units            | Value    |      |
| $h$             | nm               | 0.724    |      |
| $\tau_S^*$      | -                | 36.9     |      |
| $\kappa_{2D}^*$ | cm <sup>-2</sup> | 9.28e+13 |      |
| $f^*$           | cm <sup>2</sup>  | 2.15e-16 |      |
| $A_C^*$         | cm <sup>2</sup>  | 4.18e-14 |      |
| $C_0^{2D}$      | cm <sup>-2</sup> | 3.46e+14 |      |
| $s_F$           | -                | 3.74     |      |
| $\mathcal{F}$   | -                | 6.59     |      |
| [010] PBC       | $H_C^{2D*}$      | nm       | 1.04 |
|                 | $\phi_K$         | $kT$     | 0.75 |
| $\bar{1}10$ PBC | $H_C^{2D*}$      | nm       | 1.12 |
|                 | $\phi_K$         | $kT$     | 1.00 |
| [110] PBC       | $H_C^{2D*}$      | nm       | 1.12 |
|                 | $\phi_K$         | $kT$     | 1.00 |

Table 13 - Mechanistic parameters for Naphthalene Face {20 $\bar{1}$ }, grown from Ethanol [10]

| Face (20 $\bar{1}$ )  |                  |          |       |
|-----------------------|------------------|----------|-------|
| Parameter             | Units            | Value    |       |
| $h$                   | nm               | 0.409    |       |
| $\tau_S^*$            | -                | 16.8     |       |
| $\kappa_{2D}^*$       | cm <sup>-2</sup> | 6.22e+13 |       |
| $f^*$                 | cm <sup>2</sup>  | 4.52e-16 |       |
| $A_C^*$               | cm <sup>2</sup>  | 1.17e-14 |       |
| $C_0^{2D}$            | cm <sup>-2</sup> | 1.91e+14 |       |
| $s_F$                 | -                | 3.83     |       |
| $\mathcal{F}$         | -                | 0.95     |       |
| [010] PBC             | $H_C^{2D*}$      | nm       | 0.468 |
|                       | $\phi_K$         | $kT$     | 0.75  |
| $\bar{1}1\bar{2}$ PBC | $H_C^{2D*}$      | nm       | 0.667 |
|                       | $\phi_K$         | $kT$     | 0.31  |
| $\bar{1}1\bar{2}$ PBC | $H_C^{2D*}$      | nm       | 0.667 |
|                       | $\phi_K$         | $kT$     | 0.31  |

Table 11 - Mechanistic parameters for Naphthalene Face {110} grown from Ethanol [10]

| Face (110)            |                  |          |       |
|-----------------------|------------------|----------|-------|
| Parameter             | Units            | Value    |       |
| $h$                   | nm               | 0.450    |       |
| $\tau_S^*$            | -                | 19.4     |       |
| $\kappa_{2D}^*$       | cm <sup>-2</sup> | 6.82e+13 |       |
| $f^*$                 | cm <sup>2</sup>  | 3.92e-16 |       |
| $A_C^*$               | cm <sup>2</sup>  | 1.10e-14 |       |
| $C_0^{2D}$            | cm <sup>-2</sup> | 1.93e+14 |       |
| $s_F$                 | -                | 3.92     |       |
| $\mathcal{F}$         | -                | 0.88     |       |
| [001] PBC             | $H_C^{2D*}$      | nm       | 0.656 |
|                       | $\phi_K$         | $kT$     | 0.26  |
| $\bar{1}1\bar{2}$ PBC | $H_C^{2D*}$      | nm       | 0.692 |
|                       | $\phi_K$         | $kT$     | 0.31  |
| $\bar{1}10$ PBC       | $H_C^{2D*}$      | nm       | 0.427 |
|                       | $\phi_K$         | $kT$     | 1.00  |

Table 14 - Mechanistic parameters for Naphthalene Face {11 $\bar{1}$ }, grown from Ethanol [10]

| Face (11 $\bar{1}$ )  |                  |          |       |
|-----------------------|------------------|----------|-------|
| Parameter             | Units            | Value    |       |
| $h$                   | nm               | 0.465    |       |
| $\tau_S^*$            | -                | 21.0     |       |
| $\kappa_{2D}^*$       | cm <sup>-2</sup> | 8.32e+13 |       |
| $f^*$                 | cm <sup>2</sup>  | 3.78e-16 |       |
| $A_C^*$               | cm <sup>2</sup>  | 6.78e-15 |       |
| $C_0^{2D}$            | cm <sup>-2</sup> | 2.05e+14 |       |
| $s_F$                 | -                | 4.24     |       |
| $\mathcal{F}$         | -                | 0.52     |       |
| [101] PBC             | $H_C^{2D*}$      | nm       | 0.703 |
|                       | $\phi_K$         | $kT$     | 0.37  |
| $\bar{1}10$ PBC       | $H_C^{2D*}$      | nm       | 0.294 |
|                       | $\phi_K$         | $kT$     | 1.00  |
| $\bar{1}1\bar{2}$ PBC | $H_C^{2D*}$      | nm       | 0.551 |
|                       | $\phi_K$         | $kT$     | 0.31  |

The solubility of Naphthalene in different solvents in the temperature range 10 – 35 °C is given in Figure 12

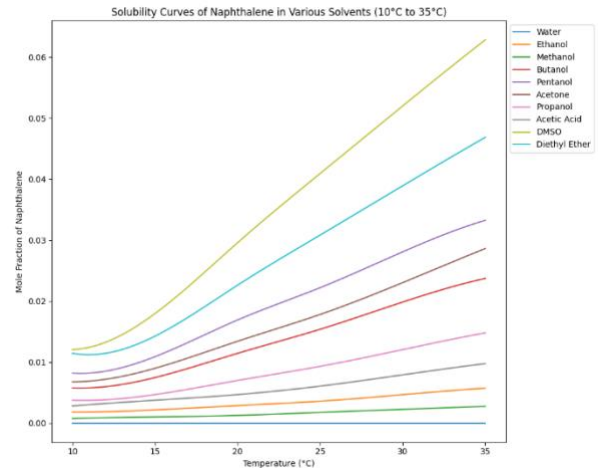


Figure 12 - Solubility of Naphthalene in different solvents